

Xavier Del Rosario

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TECHNICAL SKILLS

Programming: Java, TypeScript, JavaScript, SQL (PL/pgSQL), Python, Swift, C, C++, PHP, Bash, HTML
Frameworks / Backend: React, Vite, Node.js, Deno, Capacitor, Supabase (Postgres, Auth, RLS)
Tools / Environment: Git / GitHub, GitHub Actions, Docker, Cloudflare Pages, IntelliJ, Visual Studio, Xcode
Testing: Vitest, Playwright, JUnit, Mocha, Chai, Test Plan Development

PROJECTS

DINO — Japanese Study App (Web + iOS)

Jun 2026 – Present

- Minimized reliance on a paid translation API by serving lookups from a self-hosted 217K-entry Japanese–English dictionary in Postgres, with per-user quotas and hard spend caps on the paid fallback.
- Built a spaced-repetition review scheduler in Postgres, tuned against measured recall data so words aren't retired too early and cramming doesn't inflate scores.
- Enforced per-user data isolation with Postgres row-level security, covered by 1,600+ automated tests, including live-database integration tests, in GitHub Actions CI.
- Developed web and iOS versions from one React/Capacitor codebase (web live; iOS tested on device), writing Swift plugins for handwriting recognition, camera OCR, and photo-library access.
- Built offline review that syncs on reconnect, using server time so a changed device clock can't distort the review schedule.

Technologies: React, TypeScript, Supabase / Postgres (RLS, pgvector), Deno, Node.js, Capacitor (iOS), Cloudflare Pages

UBC Course Query — Course Project (2-person team)

Sep – Dec 2024

- Built the query engine's filtering and aggregation in TypeScript, evaluating nested AND/OR/NOT predicates with numeric comparisons and wildcard string matching, then grouping results with MAX/MIN/AVG/SUM/COUNT across 64K+ course sections.
- Converted raw zipped course files into validated, typed section records, and connected the frontend to the REST API, sending user queries and rendering results as interactive bar charts with dual-dataset overlays.

Technologies: TypeScript, Node.js, Express, Mocha, Chai

Legal Mediation Database — Course Project (3-person team)

Jul – Aug 2024

- Designed an 18-table relational schema from an ER model of a family-law mediation practice, enforcing referential integrity with 20 foreign keys, cascading deletes, and CHECK constraints.
- Built the insert, update, and delete flows in PHP over Oracle, including dependency-aware deletes that remove shared contacts, assets, and debts only when no other party still references them.

Technologies: PHP, HTML, SQL

WORK EXPERIENCE

Mill Labourer — West Fraser

Aug 2025 – Jun 2026

- Operated and troubleshoot industrial equipment in a high-throughput production line under strict safety standards.

Produce Associate — Walmart

Jul – Aug 2025

EDUCATION

B.Sc. Computer Science — University of British Columbia, Vancouver, BC

Graduated May 2025

Dean's List (2 sessions) · Cumulative average 80.3% (A-)

LANGUAGES

Japanese — Conversational